

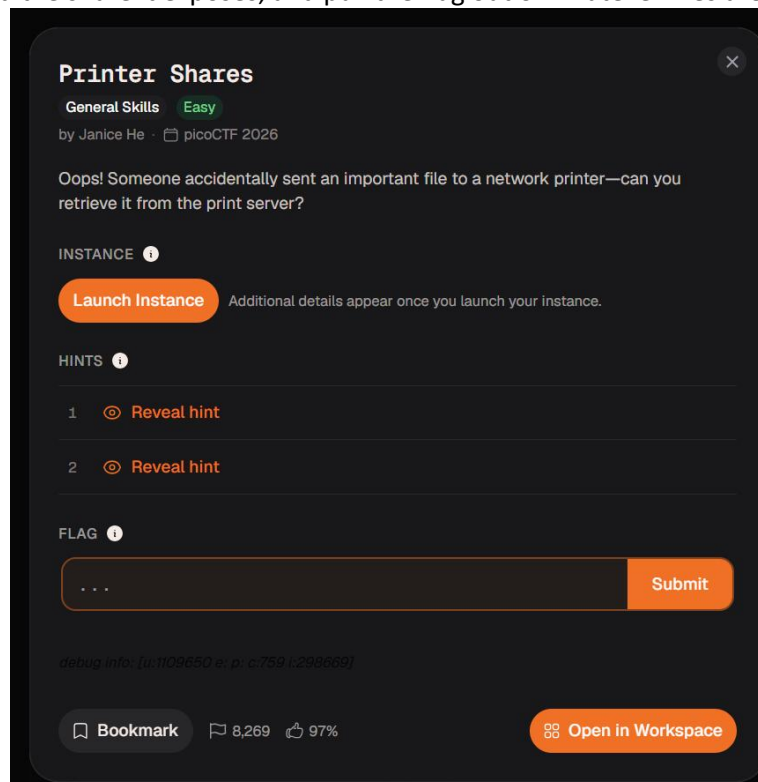
Challenge Write-Up: Printer Shares

Challenge Information

Student Name	Keyzia Joy Pascua
Section / Class	BSIT 3 - Networking
Challenge Name	Printer Shares
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 11, 2026
Flag Format Given	picoCTF{...}

1. Challenge Description

Printer Shares is a network-shares challenge built around SMB, the file-sharing protocol most commonly seen on Windows networks and served by Samba on Linux/Unix machines. The premise is that a file meant for a network printer was accidentally sent to the wrong place, and the task is to connect to that printer's SMB service, find the share it exposes, and pull the flag out of whatever files are sitting on it.



2. Initial Analysis / Reconnaissance

The challenge gives a host and a non-standard port (50545) rather than the usual SMB port 445, so the first step is simply confirming the port is open and listening with `nc -vz` before trying any SMB-aware tooling. Once connectivity is confirmed, the natural next move is `smbclient`, the standard command-line SMB client, first with the `-L` flag to list what shares the server advertises before trying to browse any of them directly.

3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
<code>nc -vz mysterious-sea.picocftf.net 50545</code>	Verify the printer's SMB port is open and reachable before connecting	Connection to port 50545 succeeded
<code>smbclient -L //mysterious-sea.picocftf.net -p 50545 -N</code>	List the SMB shares the server advertises, connecting anonymously (-N, no password)	Listed two shares: shares (Disk, public, guest access) and IPC\$ (IPC service)
<code>smbclient //mysterious-sea.picocftf.net/shares -p 50545 -N</code>	Connect directly into the shares share anonymously	Dropped into an interactive smb: \> prompt
<code>ls</code>	List the contents of the connected share	Two files: dummy.txt (1142 bytes) and flag.txt (37 bytes)
<code>get flag.txt</code>	Download flag.txt from the share to the local machine	File transferred successfully (9.0 KiloBytes/sec)
<code>cat flag.txt</code>	Read the downloaded file's contents locally	Printed the flag

4. Solution Walkthrough

Step 1

Confirm the printer's SMB port is actually open before spending time on SMB tooling: `nc -vz mysterious-sea.picocftf.net 50545`. The connection succeeds immediately, confirming there is a live TCP service on 50545 worth investigating with an SMB client.

Step 2

List the shares the server offers without yet trying to log in: `smbclient -L //mysterious-sea.picoctf.net -p 50545 -N`. The `-N` flag skips the password prompt for an anonymous (null session) login, and `-p` points at the non-standard port. The listing shows a share named `shares` described as a “Public Share With Guests”, alongside the default administrative `IPC$` share — confirming that `shares` is readable without credentials.

```
kyxxajoy@PASCUA: ~  
kyxxajoy@PASCUA:~$ nc -vz green-hill.picoctf.net 53888  
Connection to green-hill.picoctf.net (3.18.205.4) 53888 port [tcp/*] succeeded!  
kyxxajoy@PASCUA:~$ nmap -sV -p 53888 green-hill.picoctf.net  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 08:46 +0800  
Nmap scan report for green-hill.picoctf.net (3.18.205.4)  
Host is up (0.25s latency).  
rDNS record for 3.18.205.4: ec2-3-18-205-4.us-east-2.compute.amazonaws.com  
  
PORT      STATE SERVICE      VERSION  
53888/tcp open  netbios-ssn  Samba smbd 4  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 55.53 seconds  
kyxxajoy@PASCUA:~$ ipptool -tv ipp://green-hill.picoctf.net:53888/ipp/print  
get-printer-attributes.test  
Command 'ipptool' not found, but can be installed with:  
sudo apt install cups-ipp-utils # version 2.4.16-1ubuntu1.3, or  
sudo apt install ippsample      # version 0.0.0~git20220215.f365352-1build1  
kyxxajoy@PASCUA:~$ |
```

Step 3

Connect directly into that share: `smbclient //mysterious-sea.picoctf.net/shares -p 50545 -N`. This drops into an interactive `smb:` `\>` prompt. Running `ls` there lists two files, `dummy.txt` and `flag.txt`, alongside the usual `.` and `..` directory entries. `flag.txt` is a small 37-byte file, exactly the size expected for a single picoCTF flag string.

```
kyxxajoy@PASCUA: ~  
kyxxajoy@PASCUA:~$ nc -vz green-hill.picoctf.net 53888  
Connection to green-hill.picoctf.net (3.18.205.4) 53888 port [tcp/*] succeeded!  
kyxxajoy@PASCUA:~$ nmap -sV -p 53888 green-hill.picoctf.net  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 08:46 +0800  
Nmap scan report for green-hill.picoctf.net (3.18.205.4)  
Host is up (0.25s latency).  
rDNS record for 3.18.205.4: ec2-3-18-205-4.us-east-2.compute.amazonaws.com  
  
PORT      STATE SERVICE      VERSION  
53888/tcp open  netbios-ssn  Samba smbd 4  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 55.53 seconds  
kyxxajoy@PASCUA:~$ ipptool -tv ipp://green-hill.picoctf.net:53888/ipp/print  
get-printer-attributes.test  
Command 'ipptool' not found, but can be installed with:  
sudo apt install cups-ipp-utils # version 2.4.16-1ubuntu1.3, or  
sudo apt install ippsample      # version 0.0.0~git20220215.f365352-1build1
```

```

kyxxajoy@PASCUA:~$ smbclient -L //green-hill.picoctf.net -p 53888 -N

      Sharename      Type      Comment
      -----
      shares         Disk      Public Share With Guests
      secure-shares  Disk      Printer for internal usage only
      IPC$           IPC       IPC Service (Samba 4.19.5-Ubuntu)
SMB1 disabled -- no workgroup available
kyxxajoy@PASCUA:~$ smbclient //green-hill.picoctf.net/shares -p 53888 -N
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0      Tue Mar 10 05:29:06 2026
..               D           0      Tue Mar 10 05:29:06 2026
content.txt      N          1107  Thu Feb  5 05:22:17 2026
kafka.txt        N          1080  Thu Feb  5 05:22:17 2026
notification.txt N           260  Thu Feb  5 05:22:17 2026

        65536 blocks of size 1024. 57236 blocks available
smb: \> prompt off
mget *
smb: \> exit
kyxxajoy@PASCUA:~$ |

```

Additional steps (add as needed)

Download the file with `get flag.txt`, then exit the `smbclient` session and read it locally with `cat flag.txt`. This prints the flag directly to the terminal: `picoCTF{5mb_pr1nter_5h4re5_8a0df8e0}`.

5. Flag

`picoCTF{5mb_pr1nter_5h4re5_8a0df8e0}`

6. Key Concept / What I Learned

The key concept is misconfigured SMB share permissions: an SMB server that allows anonymous (null session) access to a share exposes whatever files sit on it to anyone who can reach the port, with no authentication required at all. `smbclient -L` is a fast way to enumerate what a server is willing to share before attempting to log in, and a share flagged “Public Share With Guests” is a strong signal that `-N` (anonymous login) alone will grant read access. The fix in a real deployment is to require authentication for any share holding sensitive files and to disable guest/anonymous access entirely.

7. References

None beyond the challenge's own prompt and standard documentation for `smbclient` and Samba shares.